High-Speed Link Design

Lab Report 02

PRBS Generation and Transient Analysis of a High-Speed Channel

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Cadence Virtuoso Simulation Analysis

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1 Part 1: PRBS Generation and Initialization

1.1 PRBS Generation and Verification

The PRBS generator was connected to the channel input as shown in Figure 1. The generated sequence is used as the high-speed input stimulus for the channel transient simulation.

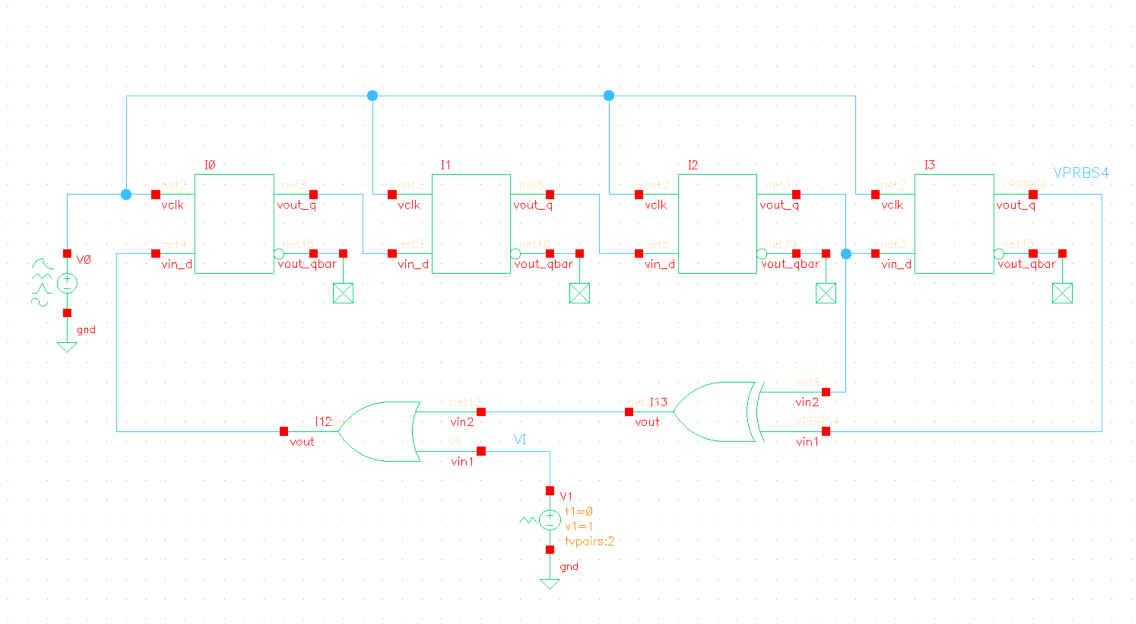

Figure 1: Schematic of the PRBS generator and its connection to the high-speed channel.

The resulting PRBS waveform obtained from the transient simulation is shown in Figure 2. The waveform switches between the two logic levels of +1 V and −1 V with a Unit Interval of 125 ps, corresponding to an 8 Gb/s data rate.

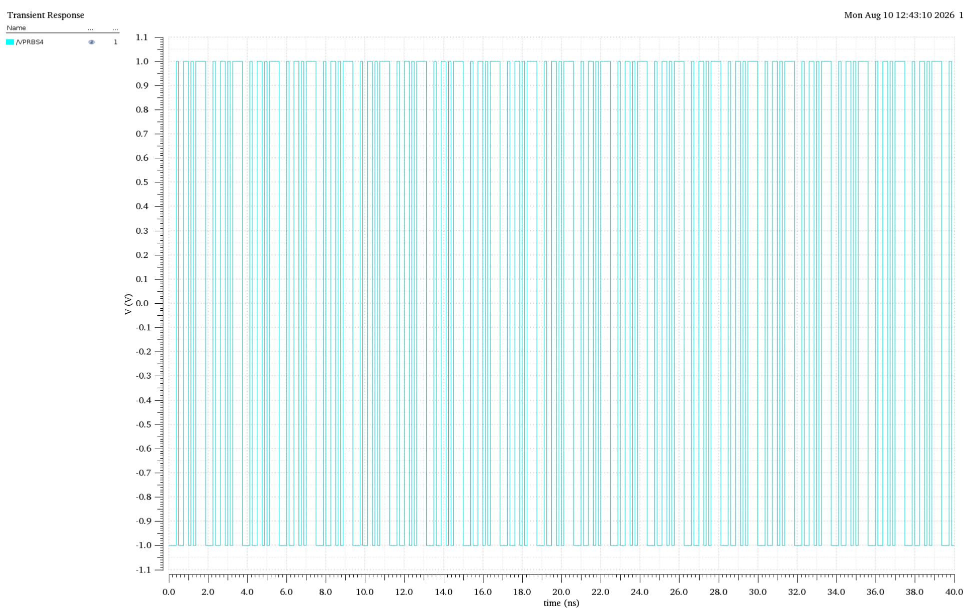

Figure 2: Transient response of the generated PRBS signal at 8 Gb/s.

To verify the generated bit sequence, an enlarged portion of the waveform was examined. As shown in Figure 3, the displayed sequence contains 15 bits. By identifying the logic levels of each bit, the sequence contains eight "1" bits and seven "0" bits:

$$N_1 = 2^n - 1 = 8, \qquad N_0 = 2^{n-1} - 1 = 7 \tag{1}$$

Therefore, the total number of observed bits is:

$$N_{\text{bits}} = N_1 + N_0 = 8 + 7 = 15 \tag{2}$$

Since each bit occupies one Unit Interval,

$$T_{\text{seq}} = N_{\text{bits}} \times \text{UI} = 15 \times 125\,\text{ps} = 1.875\,\text{ns} \tag{3}$$

which agrees with the timing markers observed in the transient simulation.

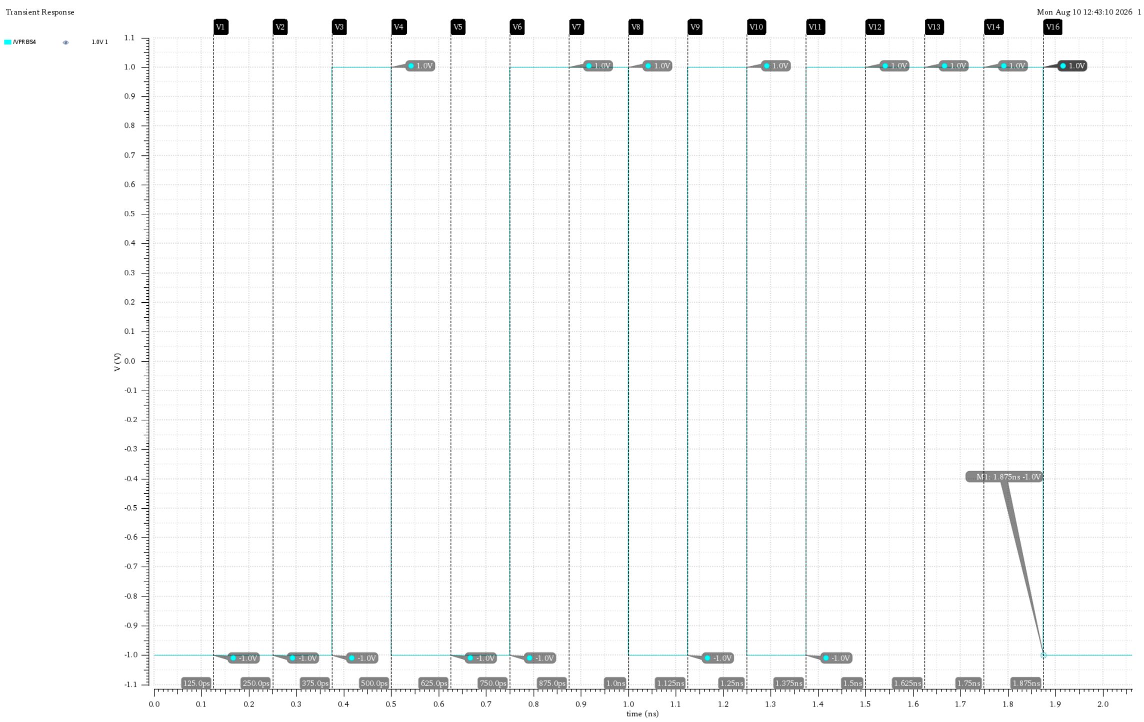

Figure 3: Enlarged 15-bit portion of the generated PRBS sequence, showing eight "1" bits and seven "0" bits.

The generated sequence was additionally checked using the corresponding PRBS verification result shown in Figure 4.

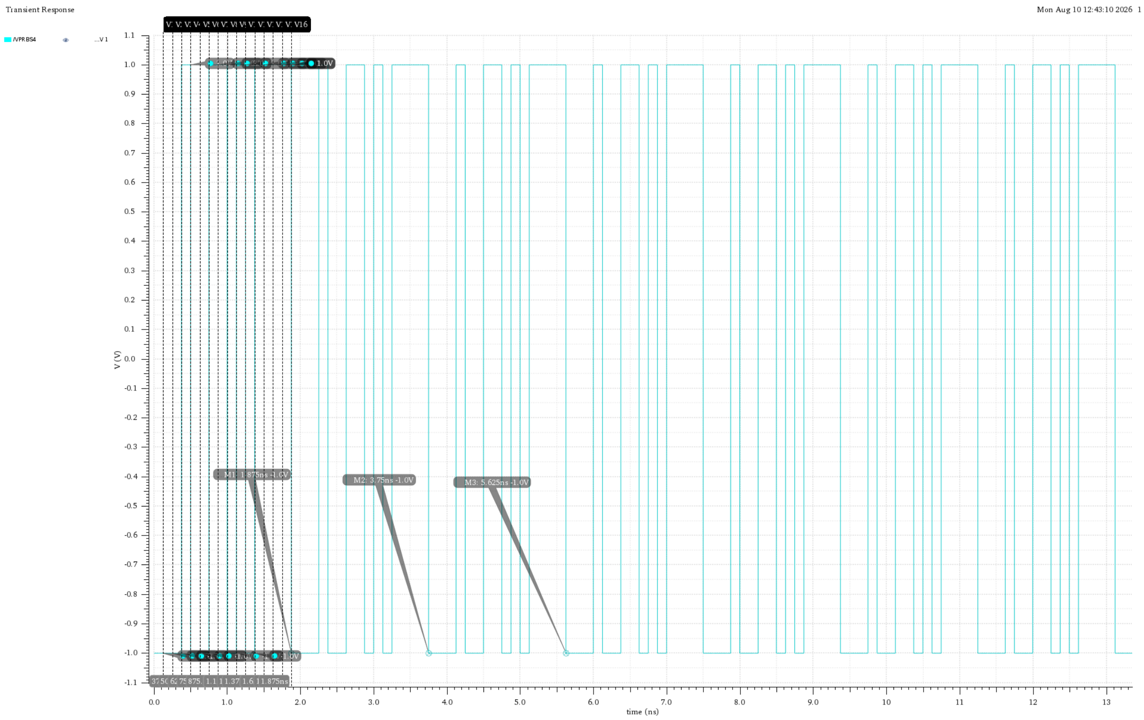

Figure 4: Verification of the generated PRBS sequence.

1.2 Comments on the PRBS Result

The transient simulation confirms that the PRBS source operates at the intended data rate of 8 Gb/s. The corresponding Unit Interval is 125 ps, which provides the fundamental timing reference for analyzing the transmitted data.

The enlarged waveform confirms the expected bit spacing and allows the individual symbols to be identified. The observed 15-bit section contains eight “1” bits and seven “0” bits, giving a total sequence duration of 1.875 ns.

It is important to distinguish the ideal PRBS source from the waveform observed after propagation through the channel. The ideal source generates well-defined binary voltage levels, whereas the channel can introduce amplitude attenuation, timing displacement, and inter-symbol interference (ISI) due to its frequency-dependent loss, phase response, and impedance discontinuities.

Therefore, the PRBS excitation provides a realistic test of the high-speed channel because it subjects the channel to consecutive transitions rather than a single isolated pulse. This makes it possible to evaluate the channel behavior under conditions that more closely represent actual high-speed digital data transmission.

1.3 PRBS Autocorrelation

The autocorrelation of the generated PRBS sequence was calculated to verify its pseudo-random characteristics. The autocorrelation measures the similarity between the original signal and a time-shifted version of itself. For a periodic signal, it is defined as

$$R_x(\tau) = \frac{K}{T_0} \int_0^{T_0} x(t)x(t+\tau)\, dt \tag{4}$$

where T_0 is the period of the PRBS sequence, $x(t)$ is the original waveform, $x(t+\tau)$ is a time-shifted version of the waveform, and K is a normalization factor. In this simulation, the normalization factor is set to

$$K = 1. \tag{5}$$

For the PRBS4 sequence, the number of bits in one complete sequence is

$$N = 2^4 - 1 = 15. \tag{6}$$

Since the bit period is

$$T_{\text{bit}} = 125\,\text{ps}, \tag{7}$$

the period of the complete PRBS sequence is

$$T_0 = NT_{\text{bit}} = 15(125\,\text{ps}) = 1.875\,\text{ns}. \tag{8}$$

Therefore, the autocorrelation was evaluated over one complete PRBS period. In Cadence Virtuoso, the autocorrelation was implemented using the following calculator expression:

$$R_x(\tau) = \frac{1}{1.875 \times 10^{-9}} \int_0^{1.875\,\text{ns}} v_{\text{prbs}}(t)v_{\text{prbs}}(t+\tau)\, dt. \tag{9}$$

In the Cadence calculator, this was implemented as:

```
integ(v("/vprbs" ?result "tran")*
lshift(v("/vprbs" ?result "tran")
pv("/tau" "value" ?result "variables"))
0 1.875n)/1.875e-9
```

The variable τ was then swept parametrically from 0 to 10 ns with a step size of 125 ps, corresponding to one Unit Interval (UI). This produces 81 simulation points and allows the autocorrelation to be observed at successive bit shifts.

1.3.1 Theoretical Autocorrelation of a PRBS

The PRBS4 used in this experiment is a maximal-length pseudo-random binary sequence, also known as an m-sequence. A fundamental property of a maximal-length PRBS is its two-level periodic autocorrelation.

For an m-stage maximal-length PRBS, the sequence length is

$$N = 2^m - 1. \tag{10}$$

When the binary sequence is represented in bipolar form,

$$1 \to +1, \qquad 0 \to -1, \tag{11}$$

its normalized periodic autocorrelation is

$$R_x(k) = \begin{cases} 1, & k = 0, \\ -\dfrac{1}{N}, & k \neq 0. \end{cases} \tag{12}$$

For the PRBS4 sequence,

$$N = 2^4 - 1 = 15, \tag{13}$$

so the expected non-zero-shift autocorrelation is

$$R_x(k) = -\frac{1}{15} \approx -0.0667, \qquad k \neq 0. \tag{14}$$

The origin of this result can also be understood directly from the number of ones and zeros in a maximal-length PRBS. A PRBS4 sequence contains

$$N_1 = \frac{N+1}{2} = 8 \tag{15}$$

ones and

$$N_0 = \frac{N-1}{2} = 7 \tag{16}$$

zeros. After converting the sequence to bipolar form, the correlation between the sequence and any non-zero cyclic shift contains one more negative product than positive product. Thus, the total correlation sum is

$$7(+1) + 8(-1) = -1. \tag{17}$$

After normalization by the sequence length $N = 15$,

$$R_x(k) = \frac{-1}{15} = -0.0667. \tag{18}$$

At zero shift, however, every element is multiplied by itself:

$$x[n]x[n] = 1, \tag{19}$$

so all 15 products are positive and

$$R_x(0) = \frac{15}{15} = 1. \tag{20}$$

Therefore, the ideal theoretical autocorrelation of the PRBS4 sequence is

$$\boxed{R_x(k) = \begin{cases} 1, & k = 0, \\ -\dfrac{1}{15}, & k \neq 0. \end{cases}} \tag{21}$$

1.3.2 Autocorrelation Simulation Result

The simulated autocorrelation is shown in Figure 5.

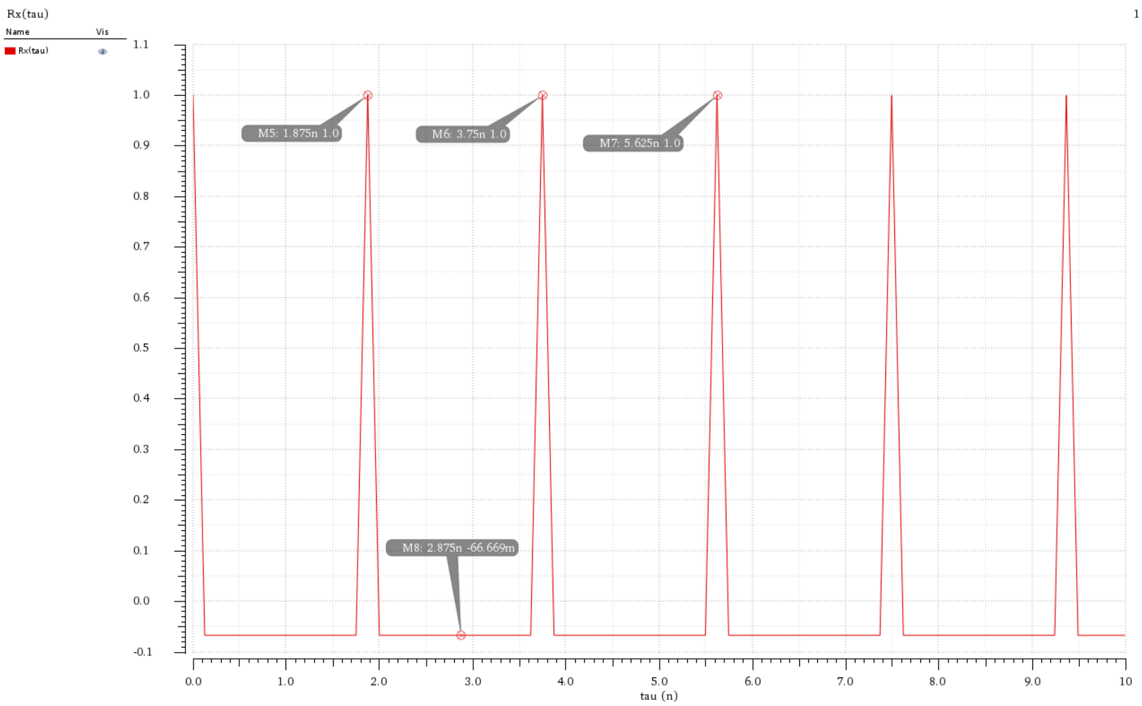

Figure 5: Simulated autocorrelation of the PRBS4 sequence for $\tau = 0$ to 10 ns.

The simulation result agrees with the theoretical behavior of a maximal-length PRBS. At zero delay, the autocorrelation reaches its maximum value of approximately

$$R_x(0) \approx 1. \tag{22}$$

For non-zero delays within one PRBS period, the autocorrelation is approximately

$$R_x(\tau) \approx -0.067, \tag{23}$$

which is very close to the theoretical value

$$-\frac{1}{15} = -0.0667. \tag{24}$$

The autocorrelation also exhibits repeated peaks at approximately

$$0,\ 1.875,\ 3.750,\ 5.625,\ 7.500,\ 9.375\,\text{ns}. \tag{25}$$

The spacing between consecutive peaks is therefore

$$\Delta\tau = 1.875\,\text{ns} = T_0. \tag{26}$$

This periodic repetition occurs because the PRBS sequence itself is periodic. After one complete sequence of 15 bits, the same sequence repeats:

$$x(t + T_0) = x(t). \tag{27}$$

Consequently, shifting the sequence by an integer multiple of its period produces the same waveform, resulting in

$$R_x(nT_0) = R_x(0) = 1, \qquad n = 0, 1, 2, \ldots \tag{28}$$

within the simulated range.

1.3.3 Comment on the Autocorrelation Result

The autocorrelation result provides a clear indication of the pseudo-random nature of the generated PRBS4 sequence. The high correlation at zero delay indicates that the waveform is identical to itself when no shift is applied, while the low negative correlation at non-zero shifts indicates that the sequence has very low similarity with its shifted versions. The value of approximately $-1/15$ is a fundamental property of a maximal-length PRBS of length 15. The repetition of the autocorrelation peak every 1.875 ns also confirms the measured PRBS period of 15 bits, with each bit occupying 125 ps. Therefore, both the time-domain waveform and its autocorrelation confirm that the generated signal has the expected characteristics of a PRBS4 sequence.

1.3.4 Characteristics of the Pseudo-Random Output

The generated output can be classified as a pseudo-random binary sequence (PRBS) based on several observed characteristics. First, the time-domain waveform exhibits a random-looking pattern of ones and zeros without an obvious short-term repetition. However, the sequence is deterministic and periodic, since it is generated using a linear feedback shift register (LFSR). For the PRBS4 used in this experiment, the sequence contains $2^4 - 1 = 15$ bits and therefore repeats every

$$T_0 = 15(125\,\text{ps}) = 1.875\,\text{ns}. \tag{29}$$

The sequence is also nearly balanced, containing 8 ones and 7 zeros. Furthermore, the autocorrelation provides strong evidence of the pseudo-random nature of the sequence. It has a maximum value of approximately 1 at zero delay and a low value of approximately $-1/15$ for non-zero shifts. The autocorrelation peak repeats every 1.875 ns, confirming the periodic nature of the sequence. Thus, the combination of a random-looking waveform, deterministic periodicity, nearly equal numbers of ones and zeros, and low non-zero-shift autocorrelation confirms the pseudo-random characteristics of the generated PRBS4 signal.

2 Eye Diagram and Pulse Response

2.1 Objective

The objective of this part is to investigate the effect of the channel on a pseudo-random binary sequence (PRBS) at an 8 Gb/s data rate. The channel response is analyzed in both the time domain and the eye diagram domain. The pulse response is used to identify the main cursor and the pre- and post-cursors, while the eye diagram is used to evaluate the resulting signal integrity and eye opening.

2.2 Simulation Setup

The PRBS signal generated in the previous part is applied to the channel under test. The data rate is

$$R_b = 8 \text{ Gb/s} \tag{30}$$

and therefore the unit interval (UI), or bit period, is

$$T_{\mathrm{UI}} = \frac{1}{R_b} = \frac{1}{8 \times 10^9} = 125 \text{ ps}. \tag{31}$$

The transient simulation was performed for a total duration of

$$T_{\mathrm{sim}} = 200 \text{ ns}. \tag{32}$$

This corresponds to approximately

$$N_{\mathrm{bits}} = \frac{200 \text{ ns}}{125 \text{ ps}} = 1600 \tag{33}$$

bit periods, providing sufficient data for constructing the eye diagram.

2.3 Simulation Schematic

The schematic used for the simulation of the channel and PRBS signal is shown in Figure 6.

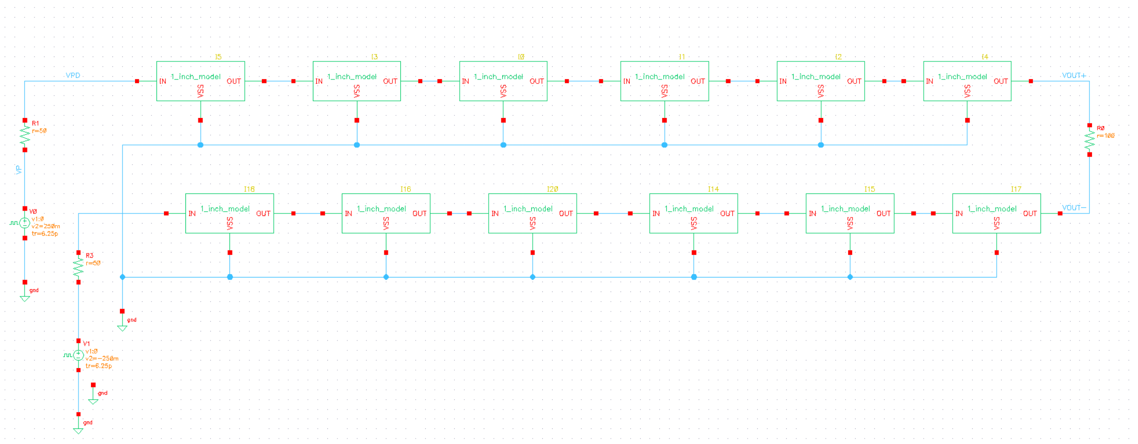

Figure 6: Cadence schematic used for the pulse-response and eye-diagram simulations.

2.4 Pulse Response

To investigate the temporal response of the channel, a single pulse with a width equal to one UI was applied to the channel. Since

$$T_{\mathrm{UI}} = 125 \text{ ps},$$

the input pulse width was therefore 125 ps.
The differential output was observed using

$$V_{\text{out,diff}} = V_{\text{OUT+}} - V_{\text{OUT}-}. \tag{34}$$

The pulse response is shown in Figure 7.

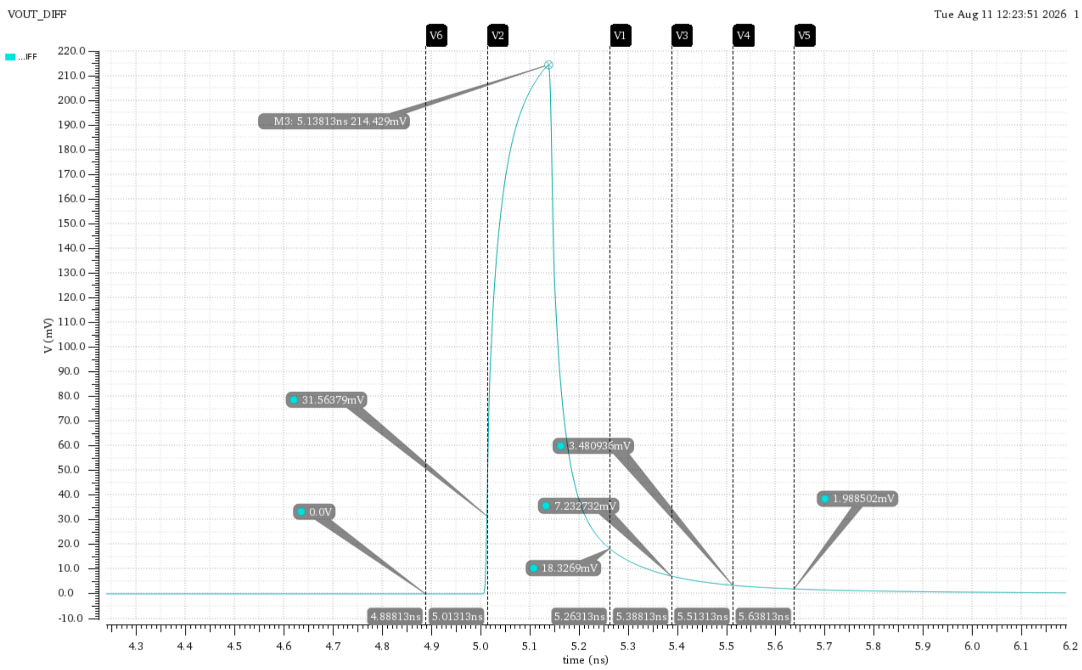

Figure 7: Simulated pulse response of the channel.

The input pulse has an amplitude of approximately 250 mV. However, the output does not reach the same amplitude within one UI due to the limited bandwidth and frequency-dependent behavior of the channel.

The main cursor of the pulse response is approximately

$$y^{(1)}(0) \approx 214.429 \text{ mV}. \tag{35}$$

Thus, the channel introduces significant attenuation compared with the input pulse amplitude.

The pulse response also extends beyond the main cursor. This indicates that the channel has memory, causing a transmitted symbol to affect neighboring symbols. These additional samples are referred to as pre-cursors and post-cursors.

2.5 Pre-Cursor and Post-Cursor Analysis

The cursor values were extracted at integer multiples of the unit interval $T_{\text{UI}} = 125$ ps relative to the main cursor.

The measured pre-cursor is approximately

$$y^{(1)}(-T_{\text{UI}}) \approx 31.56379 \text{ mV}. \tag{36}$$

The measured post-cursors are approximately

$$y^{(1)}(T_{\mathrm{UI}}) \approx 18.3269 \text{ mV}, \tag{37}$$
$$y^{(1)}(2T_{\mathrm{UI}}) \approx 7.232732 \text{ mV}, \tag{38}$$
$$y^{(1)}(3T_{\mathrm{UI}}) \approx 3.48093 \text{ mV}, \tag{39}$$
$$y^{(1)}(4T_{\mathrm{UI}}) \approx 1.988502 \text{ mV}. \tag{40}$$

The extracted cursor values are summarized in Table 1.

Table 1: Extracted pulse-response cursor values.

Cursor	Time Relative to Main Cursor	Amplitude (mV)
Pre-cursor	$-T_{\mathrm{UI}}$	31.56379
Main cursor	0	214.429
Post-cursor 1	T_{UI}	18.3269
Post-cursor 2	$2T_{\mathrm{UI}}$	7.232732
Post-cursor 3	$3T_{\mathrm{UI}}$	3.48093
Post-cursor 4	$4T_{\mathrm{UI}}$	1.988502

The presence of these pre- and post-cursors indicates inter-symbol interference (ISI). The ISI is mainly associated with the non-ideal frequency response of the channel, including its limited bandwidth and frequency-dependent attenuation. Reflections and other channel non-idealities may also contribute to the observed pulse spreading.

2.6 Eye Diagram Generation

The eye diagram was generated from the differential channel output

$$V_{\mathrm{out,diff}} = V_{\mathrm{OUT+}} - V_{\mathrm{OUT-}}. \tag{41}$$

The eye diagram was generated using the full transient simulation range from 0 to 200 ns. The trigger period was set equal to the bit period:

$$T_{\mathrm{trigger}} = 125 \text{ ps}. \tag{42}$$

For the standard one-UI eye diagram, the eye period was also set to

$$T_{\mathrm{eye}} = 125 \text{ ps}. \tag{43}$$

The eye diagram overlays consecutive one-UI portions of the received waveform. This allows the effects of ISI, timing variations, and amplitude distortion to be observed simultaneously.

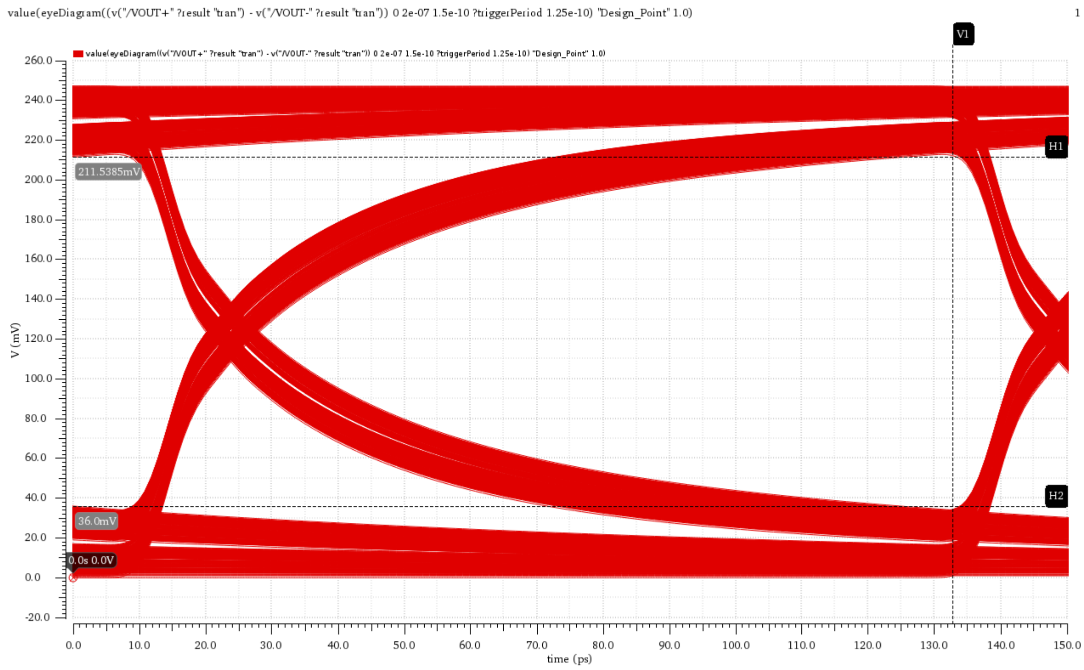

Figure 8: Simulated one-UI eye diagram of the received PRBS signal.

A multi-UI eye diagram was also generated by using a display window of 625 ps while maintaining a trigger period of 125 ps. Since

$$\frac{625 \text{ ps}}{125 \text{ ps}} = 5, \tag{44}$$

the resulting plot displays five consecutive unit intervals.

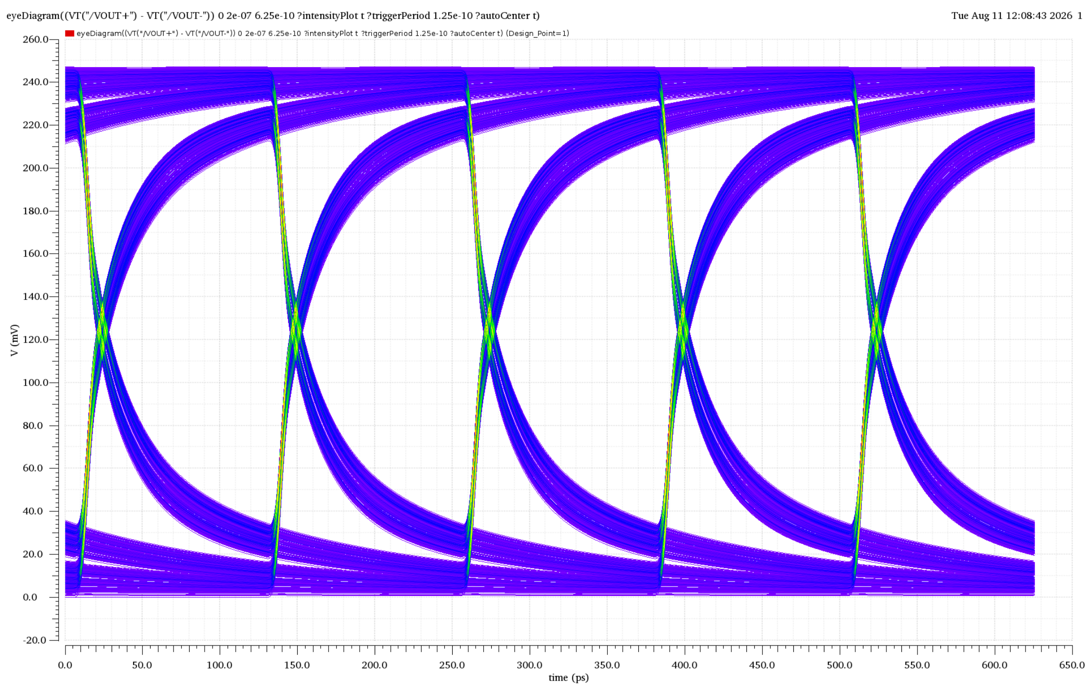

Figure 9: Five-UI display of the received PRBS waveform.

The multi-UI representation is useful for visualizing the waveform over several consecutive bits, while the one-UI eye diagram is more appropriate for quantitative eye-opening measurements.

2.7 Interpretation of the Eye Diagram

The eye diagram provides a graphical representation of the signal quality at the receiver. The vertical opening represents the available voltage margin, while the horizontal opening represents the available timing margin.

The eye is not perfectly open because the channel response extends beyond the main cursor. Consequently, the value of a received bit depends not only on the current bit but also on neighboring transmitted bits. This phenomenon is known as inter-symbol interference (ISI).

The spreading of the traces around the nominal logic levels is therefore not necessarily caused by random noise alone. In this simulation, a significant part of the spreading can be attributed to the channel's frequency-dependent response and the resulting ISI.

2.8 Worst-Case Eye Opening from the Pulse Response

The pulse response can be used to calculate the theoretical worst-case eye opening by considering the contribution of the main cursor and the pre- and post-cursors at the sampling instant.

The measured cursor values obtained from the single-pulse response are:

$$\begin{aligned} y_0 &= 214.43 \text{ mV}, && (45)\\ y_{-1} &= +31.56 \text{ mV}, && (46)\\ y_{+1} &= +18.33 \text{ mV}, && (47)\\ y_{+2} &= +7.23 \text{ mV}, && (48)\\ y_{+3} &= +3.48 \text{ mV}, && (49)\\ y_{+4} &= +1.99 \text{ mV}. && (50) \end{aligned}$$

The worst-case logic "1" level is calculated according to

$$V_{1,\text{worst}} = y_0^{(1)}(t) + \sum_{\substack{k=-\infty \\ k\neq 0}}^{\infty} y^{(1)}(t-kT)\Big|_{y(t-kT)<0}. \tag{51}$$

Since all of the measured pre- and post-cursor values are positive,

$$y_k > 0, \qquad k \neq 0, \tag{52}$$

there are no negative ISI terms. Therefore,

$$\sum_{y_k<0} y_k = 0. \tag{53}$$

Consequently, the worst-case logic "1" level is

$$V_{1,\text{worst}} = 214.43 + 0 = \boxed{214.43 \text{ mV}}. \tag{54}$$

This worst-case "1" occurs when the neighboring bits are selected such that the positive ISI contributions do not reduce the main cursor, corresponding to a sequence of the form

$$\texttt{...000010000....} \tag{55}$$

The worst-case logic "0" level is obtained from the positive ISI contributions:

$$V_{0,\text{worst}} = \sum_{\substack{k=-\infty \\ k\neq 0}}^{\infty} y^{(1)}(t-kT)\Big|_{y(t-kT)>0}. \tag{56}$$

Using the measured pre- and post-cursors,

$$\begin{aligned} V_{0,\text{worst}} &= 31.56 + 18.33 + 7.23 + 3.48 + 1.99 && (57) \\ &= \boxed{62.59 \text{ mV}}. && (58) \end{aligned}$$

This worst-case "0" occurs when the neighboring bits produce the maximum positive ISI contribution at the sampling instant, corresponding to a sequence of the form

$$\texttt{...111101111....} \tag{59}$$

Therefore, the theoretical worst-case eye opening is

$$H_{\text{eye,calc}} = V_{1,\text{worst}} - V_{0,\text{worst}}. \tag{60}$$

Substituting the measured values gives

$$\begin{aligned} H_{\text{eye,calc}} &= 214.43 - 62.59 && (61) \\ &= \boxed{151.84 \text{ mV}}. && (62) \end{aligned}$$

Thus, based on the measured single-pulse response, the calculated worst-case vertical eye opening is approximately 151.84 mV.

2.9 Comparison Between Calculated and Simulated Eye Opening

The theoretical worst-case eye opening calculated from the pulse response is

$$\boxed{H_{\text{eye,calc}} = 151.84 \text{ mV}}. \tag{63}$$

This value is compared with the eye opening measured directly from the simulated eye diagram:

$$H_{\text{eye,meas}} = \boxed{178 \text{ mV}}. \tag{64}$$

The percentage difference between the two results is calculated as

$$\%\text{Difference} = \frac{|H_{\text{eye,calc}} - H_{\text{eye,meas}}|}{H_{\text{eye,meas}}} \times 100\%. \tag{65}$$

Substituting the calculated and measured values gives

$$\%\text{Difference} = \boxed{14.6\%}. \tag{66}$$

The calculated eye opening is obtained from the individual cursor contributions of the single-pulse response, whereas the measured eye opening is obtained directly from the superposition of the actual PRBS waveform. Therefore, some difference between the two values is expected. The difference may result from the finite number of cursor terms extracted from the pulse response, the residual tail of the pulse response, and the exact sampling position used in the eye-diagram measurement.

2.10 Comment on the Eye Diagram and Pulse Response

The pulse response and eye diagram provide complementary information about the signal integrity of the channel. The pulse response identifies the main cursor and the pre- and post-cursor contributions, while the eye diagram illustrates the cumulative effect of these contributions over a large number of transmitted bits.

The main cursor is approximately 214.43 mV, compared with the approximately 250 mV input pulse amplitude, indicating attenuation introduced by the channel. In addition, the measured pre-cursor of 31.56 mV and the post-cursors of 18.33 mV, 7.23 mV, 3.48 mV, and 1.99 mV demonstrate that the channel has memory and introduces intersymbol interference (ISI).

Using these cursor values, the theoretical worst-case eye opening was calculated to be approximately

$$\boxed{151.84 \text{ mV}}.$$

This value represents the vertical eye margin after considering the worst-case combination of the neighboring bits according to the pulse-response method.